

Where α hides

Charge unit, two spheres and the reference energy

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Where α hides

Abstract

FFGFT is first and foremost a ratio-based theory. It works in natural units with the convention $\alpha = 1$ and a correspondingly redefined charge; in this basic form it needs no fractal correction. This document shows where α goes under the redefinition and what geometric meaning it has there. With $\hbar = c = \varepsilon_0 = 1$ one has $e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, and the charge is measured by the full surface of the sphere. α is then the square of the ratio of the historical to the geometric charge unit, and at the same time the ratio of two spheres around the electron, $\alpha = r_e/\lambda_C$. The FFGFT bridge $\alpha = \xi E_0^2$ reads $\xi E_0^2 = 1$ in T0 units, i.e. $E_0^2 = 1/\xi = 7500$. The only opaque element is the reference energy of the SI form $\alpha_{\text{SI}} = \xi(E_0/1 \text{ MeV})^2$: the agreement says that this reference energy is one MeV to within 10^{-4} ; a geometric interpretation of that is still lacking. The earlier reading $E_0 = 1/\xi = 7500 \text{ GeV}$ confuses E_0 with E_0^2 and attaches a unit by hand.

Status markers

- [B]** Algebraically proven.
- [K]** Numerically confirmed (check script, comparison with measured values).
- [S]** Assumption, open or speculative.
- [X]** Refuted or not sustainable.
- [Q]** Convention.

Constants: CODATA 2022. Check script: `pruef_386_alpha_geometrie.py`.

1 The basic form of FFGFT

FFGFT calculates with ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and the coupling is adopted as a unit of calculation: $\alpha = 1$ **[Q]** (A267). The only parameter is $\xi = 4/30000$. In this basic form there is no fractal correction; K_{frak} , the SI values of α and G and the fully worked-out mass tables belong to the translation into SI, not to the core (A130). The question of this document is therefore not how precisely $1/137$ is reproduced, but where α goes when it is set to 1.

2 Where α goes under the redefinition

The fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}. \quad (1)$$

In natural units with $\hbar = c = \epsilon_0 = 1$ (Heaviside–Lorentz) what remains is

$$e^2 = 4\pi\alpha \quad \textbf{[B]}. \quad (2)$$

The 4π is the full surface of the sphere. By Gauss's law the charge is the flux through a closed sphere; the flux is spread over the solid angle 4π . Setting $\alpha = 1$ gives

$$e_{\text{geo}} = \sqrt{4\pi} = 3.545, \quad (3)$$

so the charge is measured geometrically: its square is the full solid angle. With the measured α the historical charge is $e_{\text{hist}} = \sqrt{4\pi\alpha} = 0.3028$. Hence

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}} \right)^2, \quad \frac{e_{\text{geo}}}{e_{\text{hist}}} = \frac{1}{\sqrt{\alpha}} = 11.706 \quad \textbf{[B]}. \quad (4)$$

Under the redefinition α therefore does not disappear. It moves into the charge unit: α is the square of the ratio of the historical to the geometric charge unit. In every calculation it stays on record through $e^2 = 4\pi\alpha$, one factor $\sqrt{\alpha}$ per vertex (A267).

The same 4π appears in the Higgs–vacuum route: the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ of the 2025 vacuum formula is the 4π from $4\pi\epsilon_0$ (Doc. 385).

3 The geometric picture: two spheres around the electron

Two spheres can be drawn around the electron. On the first, the Coulomb energy equals the rest energy; this is the classical electron radius

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} = 2.818 \times 10^{-15} \text{ m.} \quad (5)$$

On the second, the quantum energy $\hbar c/r$ equals the rest energy; this is the reduced Compton wavelength

$$\lambda_C = \frac{\hbar}{m_e c} = 3.862 \times 10^{-13} \text{ m.} \quad (6)$$

Their ratio is exactly the fine-structure constant:

$$\frac{r_e}{\lambda_C} = \alpha, \quad \frac{a_0}{\lambda_C} = \frac{1}{\alpha} \quad [\mathbf{B}]. \quad (7)$$

Equivalently, α is the ratio of Coulomb energy to quantum energy on the same sphere, $(e^2/(4\pi\epsilon_0 r))/(\hbar c/r)$, for every radius r **[K]**.

Geometrically, $\alpha = 1$ means that the Coulomb sphere and the Compton sphere coincide, and with them the Bohr radius; r_e , λ_C and a_0 are then a single scale $1/m_e$ (A267). The SI value $1/137$ only says that in the historical charge unit the Coulomb sphere is 137 times smaller than the Compton sphere. This is the transparent geometric meaning of α .

4 The FFGFT bridge in ratio form

FFGFT links α to ξ through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons. In the basic form, i.e. in T0 units with $\alpha = 1$ and without fractal correction, the bridge reads

$$\xi E_0^2 = 1 \implies E_0^2 = \frac{1}{\xi} = 7500, \quad E_0 = 86.60 \quad [\mathbf{B}]. \quad (8)$$

The number $7500 = 1/\xi$ therefore belongs to E_0^2 , not to E_0 . Taken by itself, this equation fixes the T0 energy unit: one unit is $E_0/86.60 = \sqrt{\xi m_e m_\mu} = 84.85 \text{ keV}$ **[K]**. It acquires content only by comparison with a second scale, and that is the transition to SI.

5 The SI bridge and the reference energy

In SI the ratio form becomes (Doc. 011, A130)

$$\alpha_{\text{SI}} = \xi \left(\frac{E_0}{E_{\text{ref}}} \right)^2, \quad E_{\text{ref}} = 1 \text{ MeV.} \quad (9)$$

This is where the bridge becomes opaque. ξE_0^2 has the dimension energy², so a reference energy is needed, and the corpus takes the MeV. Computing backwards which reference energy the equation requires with the measured α gives

$$E_{\text{ref}} = E_0 \sqrt{\xi/\alpha} = \begin{cases} 0.99323 \text{ MeV} & \text{bare, } E_0 = 7.348 \text{ MeV,} \\ 0.99992 \text{ MeV} & \text{corrected, } E_0 = 7.397 \text{ MeV} \end{cases} \quad [\mathbf{K}]. \quad (10)$$

The bare value is $\sqrt{K_{\text{frak}}}$ MeV to within 8×10^{-5} . The fractal correction therefore enters only here, as a fine adjustment between the T0 unit and the MeV; without it $\alpha^{-1} = 7500/E_0^2 = 138.9$,

with it 137.06 , 1.7×10^{-4} from the measured value. The same in the units of the basic form: the T0 energy unit is $\sqrt{\alpha} \cdot E_{\text{ref}}$.

The agreement of α thus states one thing: the reference energy is one MeV to within 10^{-4} . This is a property of the anchor E_0 (R135), not an independent prediction **[K]**. A geometric meaning of $E_{\text{ref}} = 1$ MeV is open **[S]**. Obvious candidates do not work: the pair threshold $2m_e = 1.022$ MeV is off by 2.2 % **[X]**. The statement in Docs. 013 and 014 that $S_{T0} = 1$ MeV/ c^2 is predicted is circular, because $m_e^{T0} = 0.511$ is already given in MeV; Doc. 014 carries a note on this, and Doc. 013 now does too.

6 Consequence for the reading $E_0 = 1/\xi$

Docs. 005 and 032 contain $E_0 = 1/\xi = 7500$ GeV; Docs. 133 and 165 took the equation over. The ratio form shows what is right about it and what is not. The number is right: in T0 units $E_0^2 = 1/\xi = 7500$ **[B]**. Two steps are wrong **[X]**. First, the square is missing: with $E_0 = 1/\xi$ one would have $\xi E_0^2 = 7500$, neither 1 nor α . Second, the unit GeV is attached to a pure number; 7500 GeV lies a factor 1.01×10^6 above $E_0 = 7.397$ MeV.

7 What holds and what is open

Statement	Status	Reason
$e^2 = 4\pi\alpha$, with $\alpha = 1$, $e = \sqrt{4\pi}$	[B]	$\hbar = c = \epsilon_0 = 1$
$\alpha = (e_{\text{hist}}/e_{\text{geo}})^2$	[B]	redefinition of the charge
$\alpha = r_e/\lambda_C$, $a_0/\lambda_C = 1/\alpha$	[B]	two spheres around the electron
$\alpha = 1$ as unit of calculation	[Q]	A267
$\xi E_0^2 = 1$, $E_0^2 = 1/\xi = 7500$ in T0 units	[B]	basic form
T0 energy unit $\sqrt{\xi m_e m_\mu} = 84.85$ keV	[K]	check script
$E_{\text{ref}} = 0.99992$ MeV (corrected), 0.99323 MeV (bare)	[K]	property of the anchor (R135)
geometric meaning of $E_{\text{ref}} = 1$ MeV	[S]	open
$E_{\text{ref}} = 2m_e$	[X]	off by 2.2 %
$S_{T0} = 1$ MeV/ c^2 predicted	[X]	circular
$E_0 = 1/\xi = 7500$ GeV	[X]	square missing, unit set by hand

8 Aligned places in the corpus

With this document the places with $E_0 = 1/\xi$ carry a dated note pointing to the ratio form $E_0^2 = 1/\xi$: Doc. 005 (previously without a note), 032, 133 and 165, each De/En. Doc. 384 puts the basic form first. Doc. 013 notes that $S_{T0} = 1$ MeV/ c^2 holds by construction.

9 Conclusion

α does not disappear under the redefinition; it becomes the charge unit. With $\alpha = 1$ the charge is measured by the full surface of the sphere, $e = \sqrt{4\pi}$, and the Coulomb sphere of the electron coincides with its Compton sphere. The measured value $1/137$ is the radius ratio r_e/λ_C in the historical charge unit. In its basic form the FFGFT bridge reads $\xi E_0^2 = 1$, i.e. $E_0^2 = 1/\xi$; this is transparent. Only the step to SI is opaque, because it needs a reference energy that is one MeV to within 10^{-4} , without a geometric reason for it being available.

Check script: 2/python/Dok386_Skripte/pruef_386_alpha_geometrie.py — 19/19.